

24-09-24
CS-222

Analysis + design of software system

System

A system is a integrated whole.

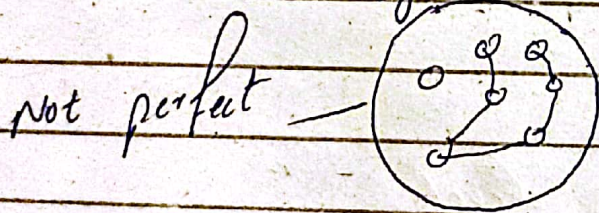
(Two system can be dependent on each other like classes depends on transport system.)

Integration

It means that components are connected with each other.

Perfect system

The perfect is a system in which all part are integrated with each other.



Design

Every system has design that is naturally or by 3rd party.

Analysis

- Identification of components
- Application / use
- How it used, Limitation.

Browser is not currently use as application software.



01-10-24
Again October

Software Development

- It is not only sketch but also execution of software.
- It is a large scale.

Attributes of Good software

- 1 - Safety —
- 2 - Reliability —
- 3 - Efficiency —
- 4 - Maintainability
- Acceptability -

Safety and Security Diff

— Safety has high level
Theif in house and you lock the door
your security is done but safety is not

Scalability:-

To increase capacity.

Role of Law

If there is no law people will not respect the each other.

Application

- 1 stand-alone App - (That work alone calculator)
- 2 Distributed App - (Playing Ludo with friend)
- 3 transaction based App - (Like NBP, EasyPassa)
- 4 Embedded control App - (Firmware, EROM, these memory used in system the app, like open car by remote)
It is used for dedicated or small task.
(like only for one task.)

5) Batch Processing system:-

Multiple task can work in Batch.
Batch is a box of command that work in once and the task performs sequentially.
If we have five task we do one after another is Batch.

ERP

~~Enter Price~~

Enterprise Resource Planning

Ref: Software Eng Code of Ethics and Professionalism

Software code also a data.

Confidence

It means that you have ability and try to attain confidence

Intellectual Property Rights

Make open source your software.

Computer Misuse

Don't use

Competence

Case study

An embed system

It will be a software that control hardware.

Issues: — physical ^{size} ~~size~~, power management

Example: A software that control pump of insulin for the diabetes patient.

2. Information System

Able to add, delete,
Restore, update, Backup

The system is used to manage and provide access to database of information.

issues: security, usability, privacy
Example: Medical record system

3. A sensor Based data collection system

To collect data from diff sensors and manage in a way.

issues: reliability, environment, maintainability
Example: A wilderness weather station.

4. Support Environment

This system is used to support a certain activity.

issues - task coverage, easy learn
Example: Visual studio, like this environment support us to do programming in a good way.

Software Process

A set of related activities that related leads to the production system.

Fundamental software ^{engineering} activities:

These activities are common to all software

1 software specification

2 software development

3 software validation

What the ~~to~~ customer wants?

4 software evolution

To meet changing ~~to~~ customer needs.

Some other Activities

1 software configuration management

2 Project Planning

Some subactivity

1 Requirement Validation

You can use survey for this purpose

2 Architectural design

3 Unit testing

When Describe Process:-

It is important who is involved, what produce, and condition that disturb the process.

1. Products :-

they are the outcomes of activity.

2. Roles

People Roles like manager, programmer, designer.

3. Pre, Post Condition

Condition that must hold before and after the activity.

Types of Process

1. Plan driven process

The activities that are upper mention

is called plan driven process.

It is used for safety safety purpose.

2. Agile processes

It is process where planning is incremental and continual as the software is developed. It is easy when requirements are changed by user. where small changing reqi

SDLC (Software Development Life Cycle)

The Water fall model:

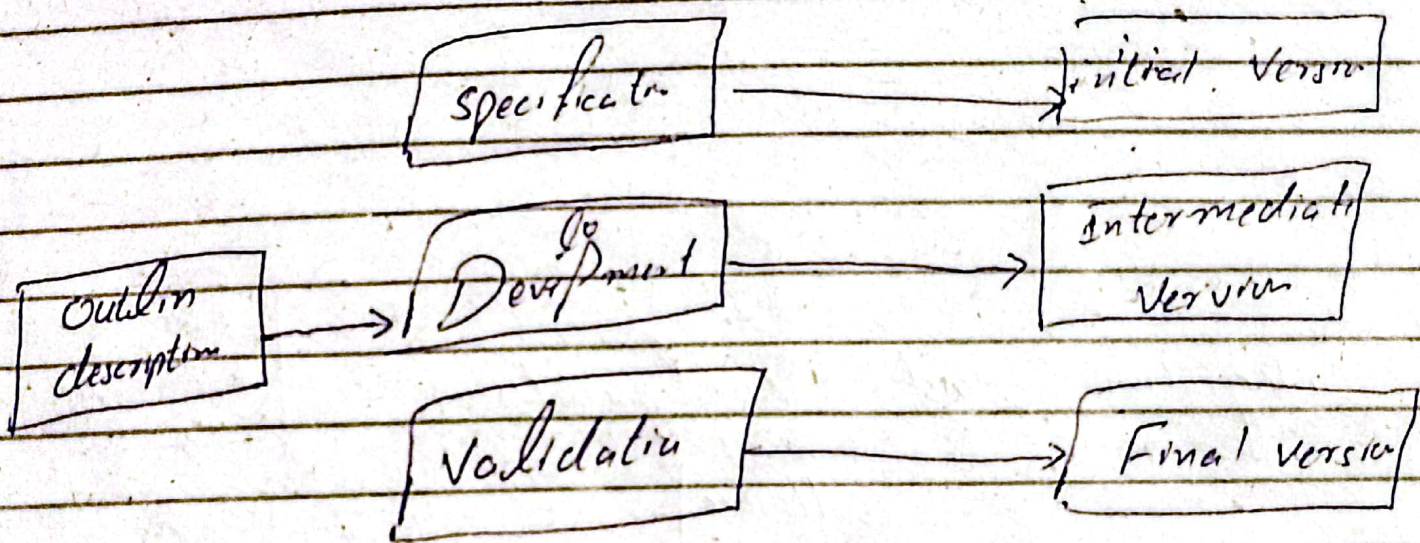
In this model we can't go back

The incremental Model

10-10-24

Incremental development

- When the model is not error free then this should be used.
- No complete requirements are necessary.
- It just need a short description.



Problem:

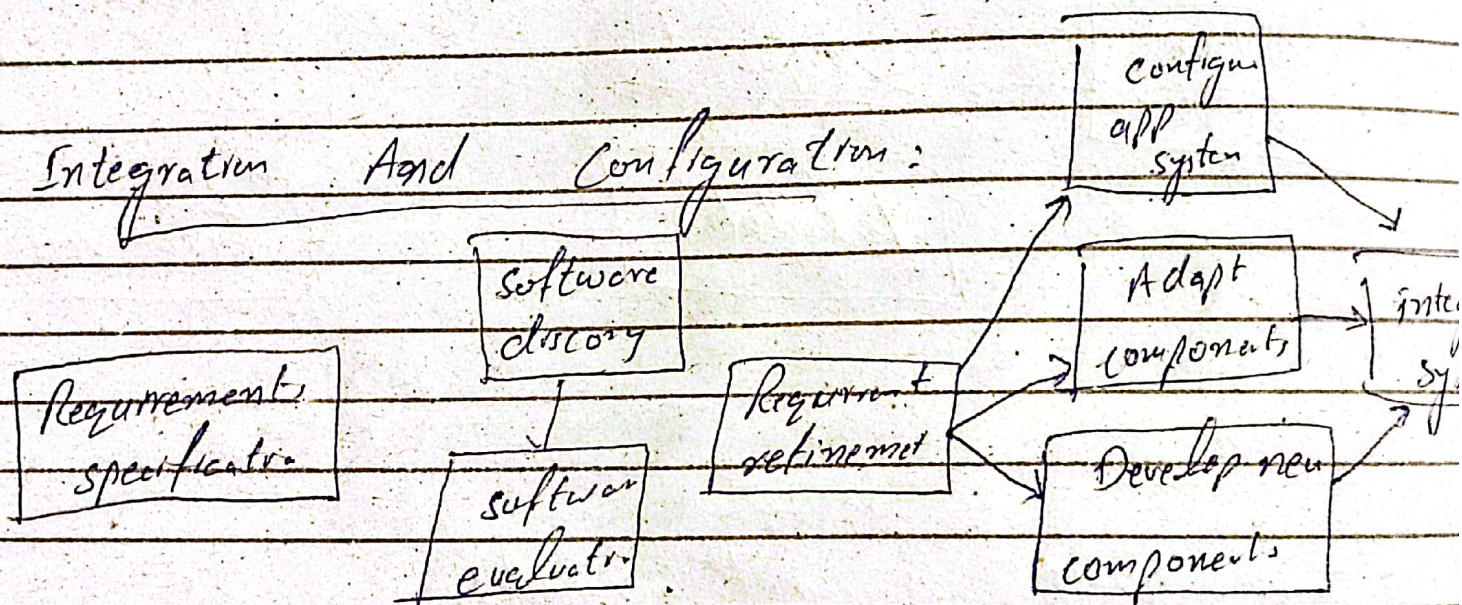
The process is not viable:

system structure tends to degrade as new increments are added:

Spiral Development

- Create the plan of requirement
- Review.
- Make Prototype
- Requirement validation (It is only in document for)
- Open source

Integration And Configuration:



My App

It is the combination of two app to fulfill the user need.

~~DATA~~ MAP API

It is the MAP in which all maps are visible. It is used by cream indrive etc.

Components frequently used:

1 stand alone app system :

2 Collections of object

3 Web services :

It is not downloaded but used for a particular purpose. Like you can check your sim cards by entering your Id card no.

Process Activities :

* Requirement Evaluation:

You must full full fill the user requirements. software requirement is image of user requirement.

7.1 Requirement elicitation and analysis:

2) Requirement Specification
System must be user friendly. It
is more efficient than previous version.

3) Requirement Validation :



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Software Design And Implementation:-

Design Inputs

- Platform information
- Software requirements (SRS (software requirement specification))
- Data description (which kind of data will use)
example: like instagram has picture and video.



Design Activity

- Architectural Design
- Interface Design
- Database Design
- components selection and design.

Design Output

- System Architecture
- interface
- Database
- components selection

Software Validation:-

Component testing

System Testing

Customer testing

Software Evaluation

Software engineering is an evolutionary process where software is continually changed over its lifetime in response to changing requirements and customer needs.



Assignment

Try to identify three requirements of user of any app or software according to your choice.

For each user requirement write the system requirements.

~~Signature~~
~~Date~~

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How we capture requirement?

1 - Interviews

- open intervals interview } domain is fixed
- Focus interview / structure interview } in both

Fixed set of
questi.

- i) How user will use machine?
- ii) How long he will use?

2 - Survey

You can conduct survey.

- Questions (You can ask question from stakeholders)

Example: (Fill form for Psychology issues)

- Now it can be conducted online via e-mail or on other tools.

3) Ethnography

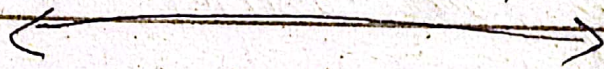
- Watching and observing the system for a specific time period.
- It also used to know who people will use system (users).

4) Scenario

- It is a user story.
- A detail description of user interaction.
- Scenario based design of system.
- It could have multiple scenarios as goes hand in hand with future based development.

- Scenario of interacting with ATM Machine

calls work
(Last time I when I went to a ATM machine of university But unfortunately it was not working then I went to an other ATM machine near my house but it also not working. At last I reached at a market I had wait for 15 min to enter in machine. Firstly I was happy that it is working the AC was at 22°C. the machine light was on card inserted light was blinking and I put card in it. Machine accepted the card. I ~~insert~~ entered the pin then select the amount. Happy sound I hear that the money is collecting. Then money come out. Experience was not good. As I have to wait ~~and~~ for my turn.)



Scenario - must be engaging.

Actors: (Like student, teacher, manager of Bank)

They are actors of ATM.

Resources

You should know what resources are necessary.

Requirements:

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Dependability

→ Availability

Structured Information

→ A description of function being specified
→ A description " " inputs and
→ ~~information~~ about " & their sources.

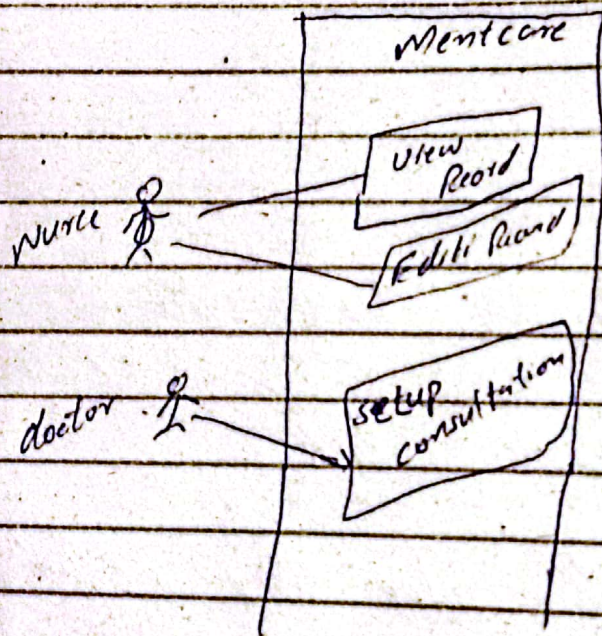
→ information about the information needed
for the computation entities in the
system that are required.

→ A description of action to be
take

→ If a functional approach is used, a
precondition setting out what must
be true before function is called
and a post condition what will
be output.

→ A description of side effect.

Problem:- Some time you can't specified
the requirement by using natural language.
So it can be expressed in a
table model.



05-11-24

Graphical Notation to Present

User Requirements.

UML

Unified Modeling Language.

Case Tools:

Computer Aided Software Engineering:-

Each phase has input that give output that used as input for next phase. They are called automated code generator.

Automated code Generator

Automated test Generation

Types of test to accept or reject

→ Acceptance test (It is normally test by user)

→ Unit test

For these test we use test case to verify them

Use Case / Use case description

→ Draw use case diagram

→ Actors (Human, Non, Human)

Non, Human (It can be ^{another} subsystem or sub system)

07-11-24

Actors:—

Teacher, Student, CR

Identify the Actors

- i) Assign Assignment
- ii) Review Assignment
- iii) Submit Assignment
- iv) Update Assignment
- v) Coordinate with teacher.
- vi) Route

Make Use Case:—

Question

~~Assign Assignment~~ Submission of Assignment

Teacher Assign the Assignment to the student with a dead line individually or on group basis.

Review Assignment

Teacher has a quick ~~review~~ review of assignment to check wheather it is ok for submission

Submit Assignment

Student has to submit assignment before the deadline

Update Assignment:

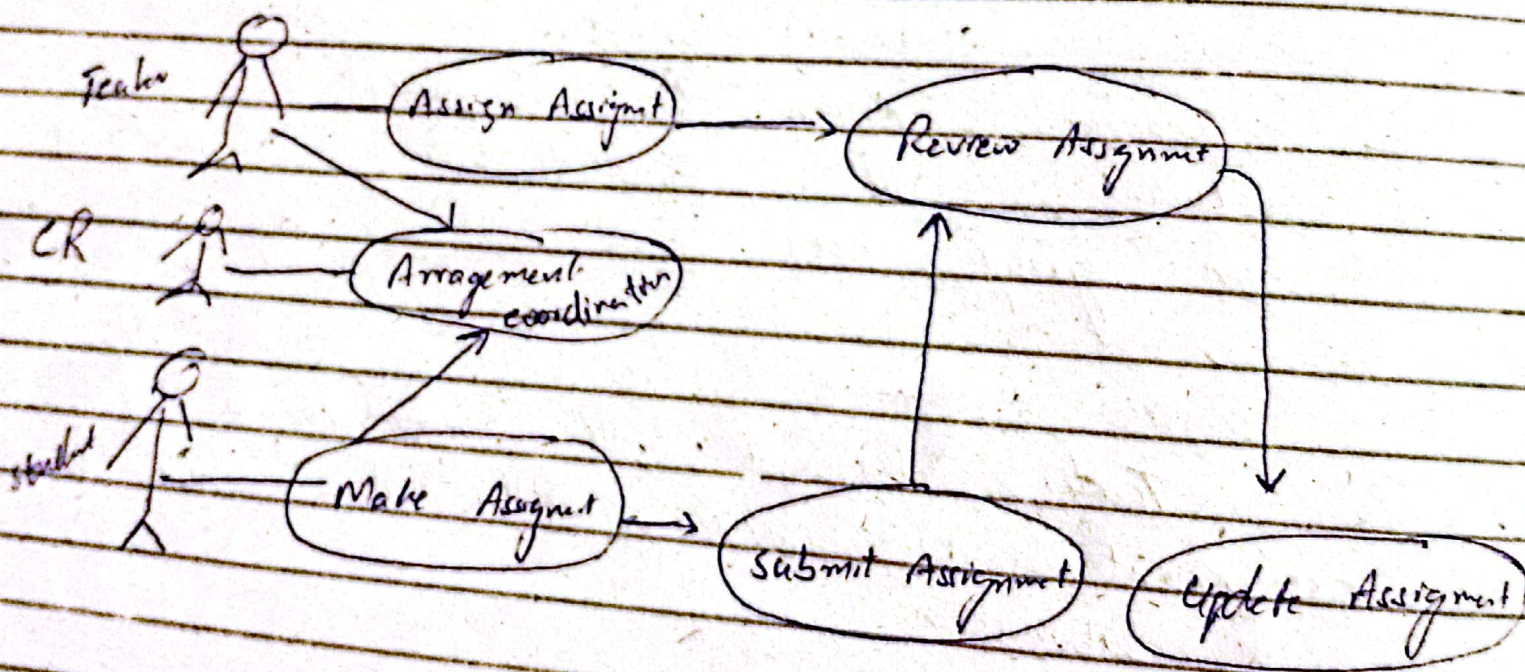
Update the assignment to submit again
you can do it only for once

Resubmit Assignment:

Resubmit the assignment after the
updatation.

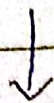
Coordinate with teacher:

CR coordinates with teacher to
arrange an logistic arrangement with whole
class.



12-11-24

Use Case Diagram
UCD

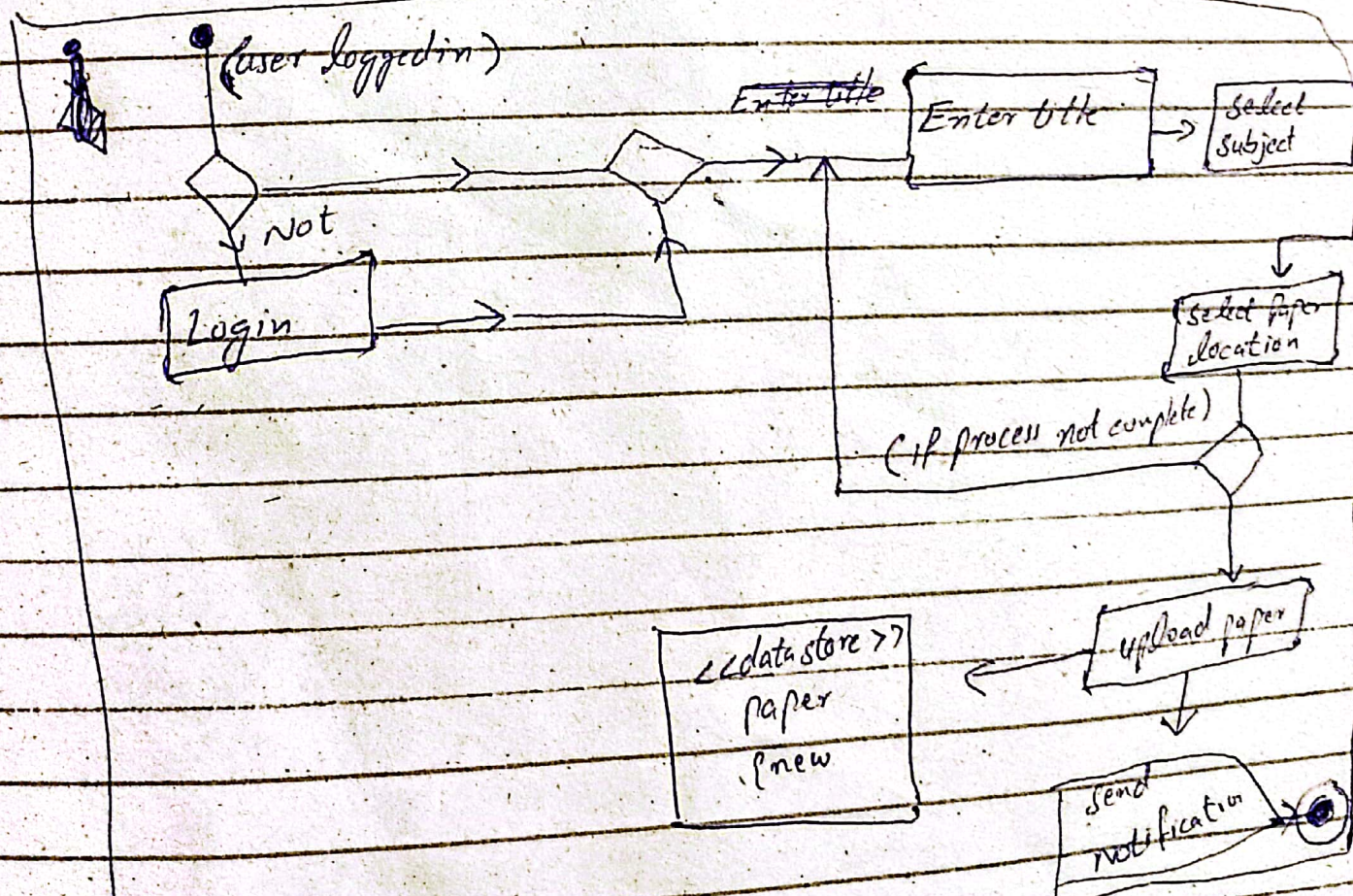


Activity Diagram
AD

In Use case we have many activities and also have diff activity diagram for each activity.

Conference Reviewing System :-

→ Submission Process



ADSS

System Modeling:-

Process of developing of a system with each model presenting a different view or perspective of that system.

Model is not an exact system.

Four types of Model

1. Content Model

External perspective

2. Interaction Model

Interaction b/w a system and its environment or b/w the components of system.

3) Structural Model

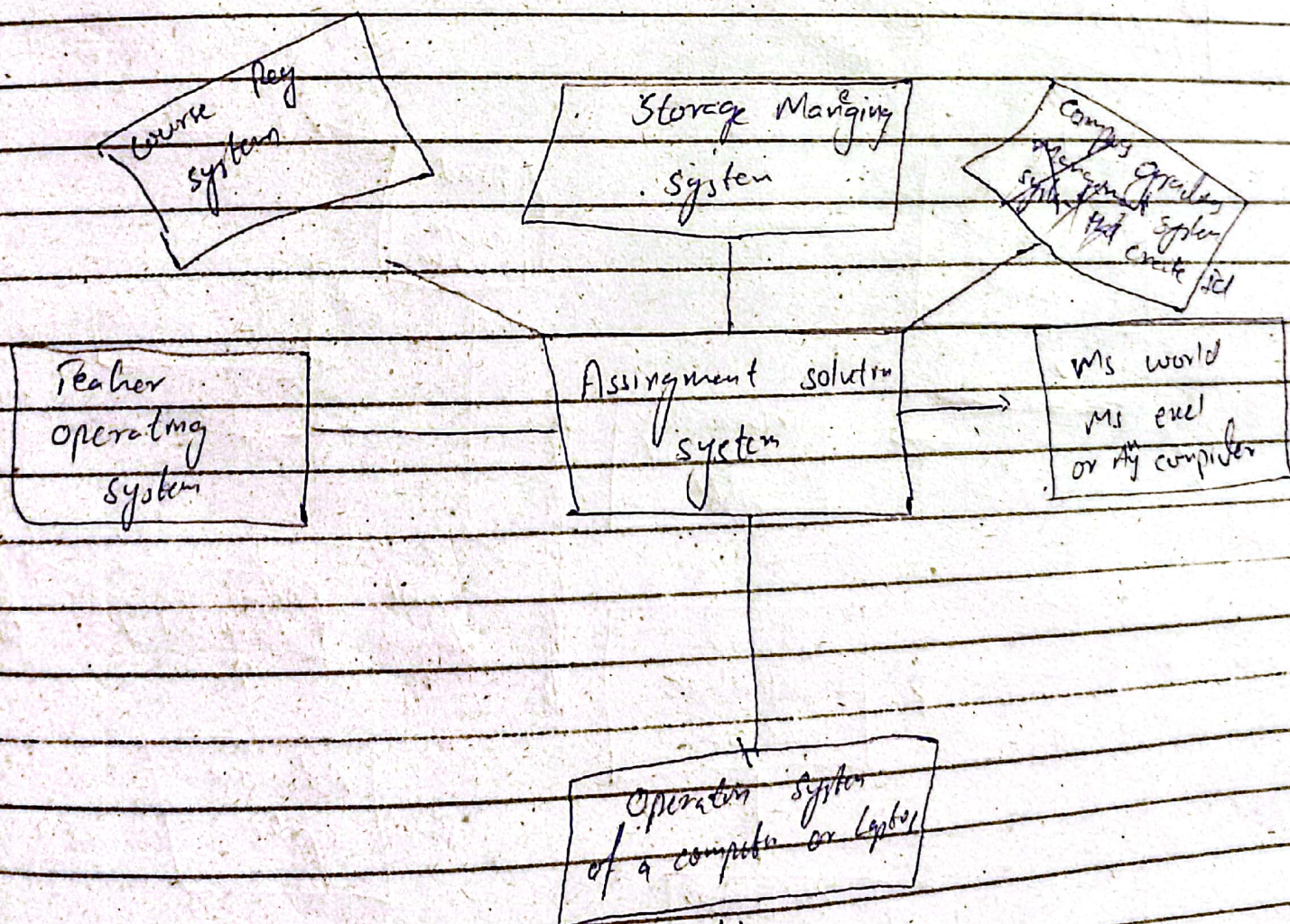
Organization of a system or structure of the data.

4) Behavioral Model

Dynamic behavior of the system and how it responds to events.

Representing Graphically
 We can use this language
Unified Modeling Language.

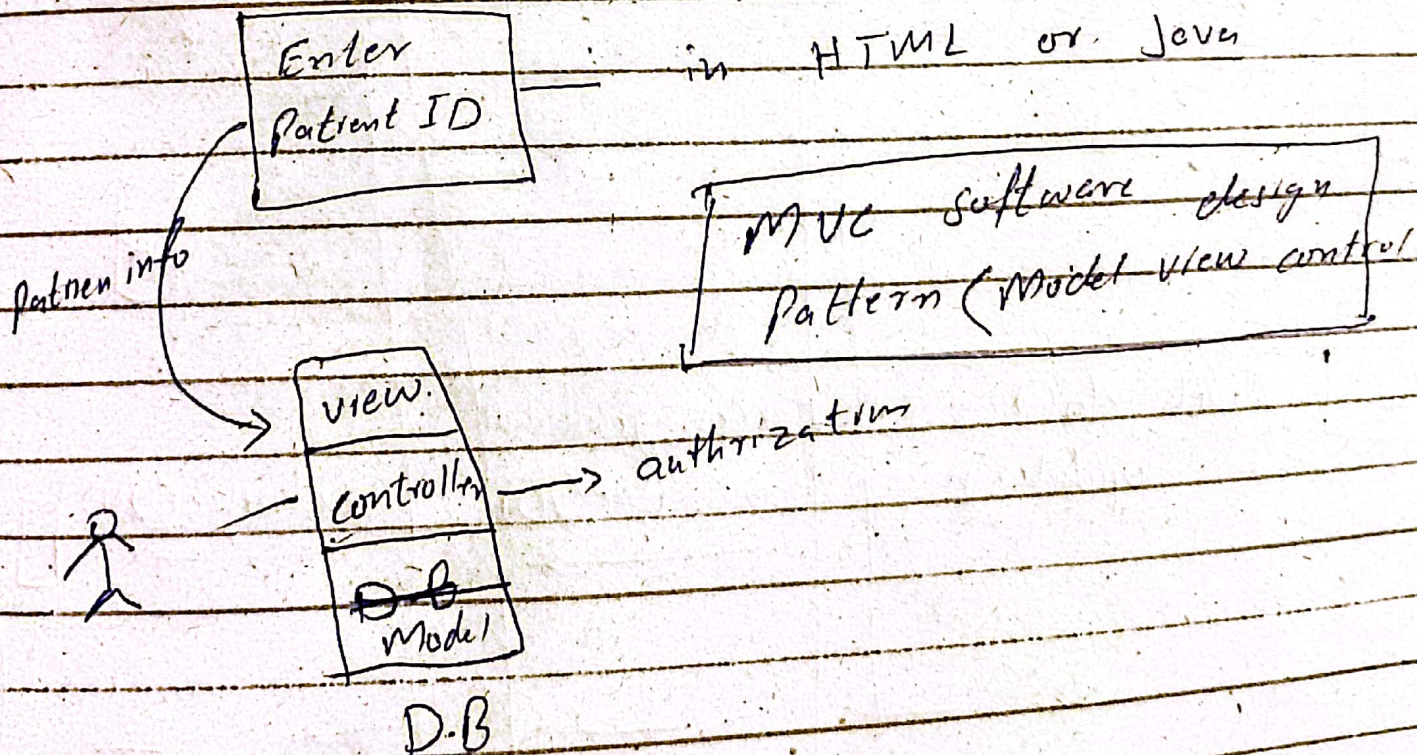
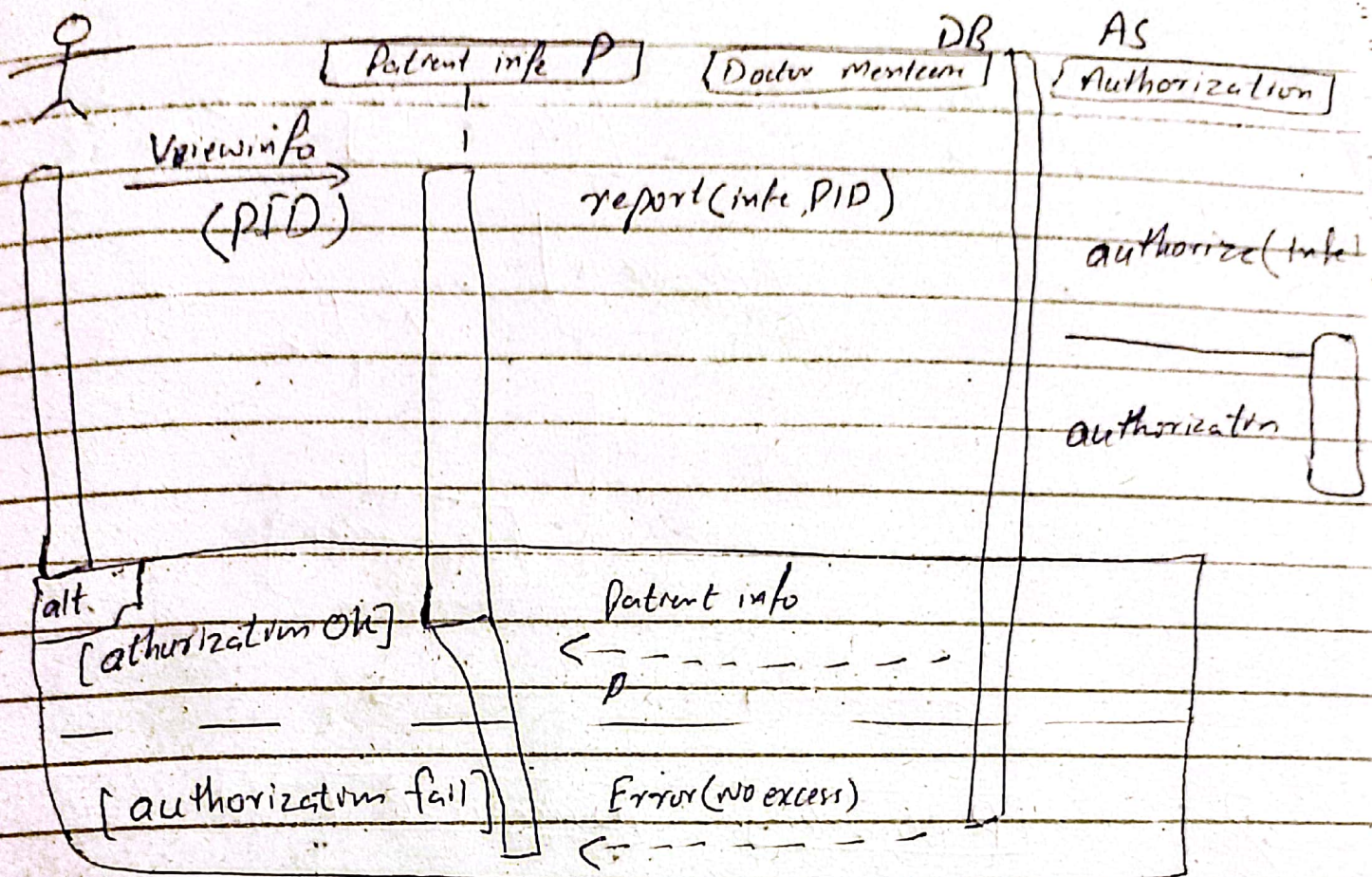
- a) Activity (shows activity involve in process
- b) Use case diagram (shows the interaction b/w system and environment
- c) class diagram : show the object classes in the system and association b/w the classes.
- e) state diagram : show how system reacts to internal and external events.



Context Model

21-11-2

Sequence Diagram



03-12-21

Structural Model

Structural Model

Display the organization of a system in terms of the components that make up that system

→ Classes

Entitled with common characteristics.

→ Association

Relation of classes with each other is called association.

First class Entity

Entity that can be managed by its own.

$[1^*]$ → It means that one must

Cardinality

It will tell the direction

Arity

It will tell the number of association

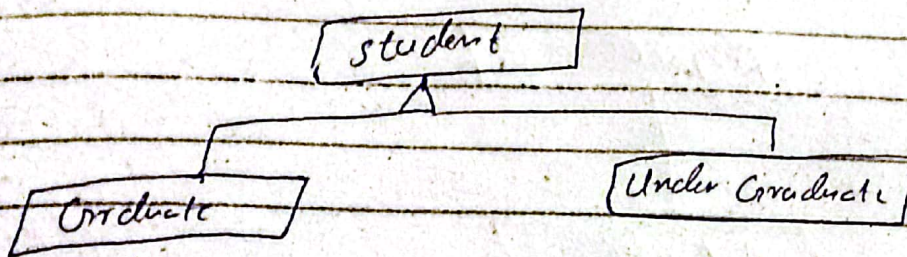
Patient
Name
age
Gender
disease
new()
prescribe()
treatment()
meditation()

Doctors
Name
Doctor type
new patient()
give prescribe,
check
patient()
give advice()

Hospital
Name
Address
Type
Rooms
Admission()
Operation()

Meditation
Medicines
Therapies
Doctor
type
new()
prescribe()
treatment()

Generalization Hierarchy

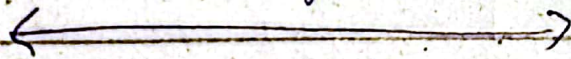


→ sub class inherited from super class and also inherit some other properties and operations.

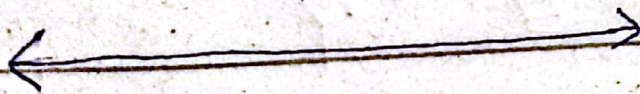
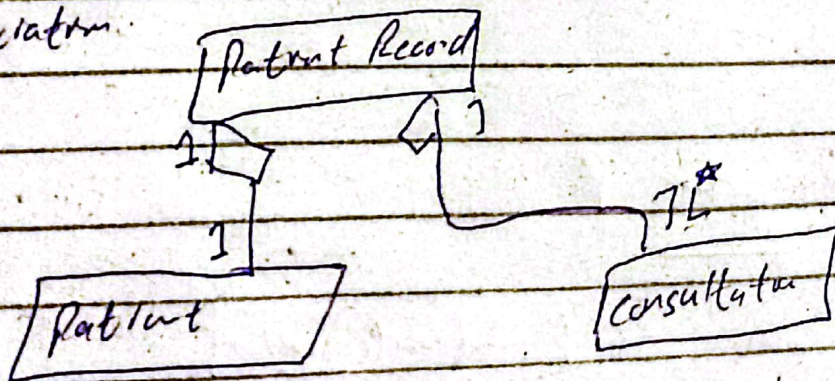
→ class should always have some member.

Example:- Like ~~to~~ table class

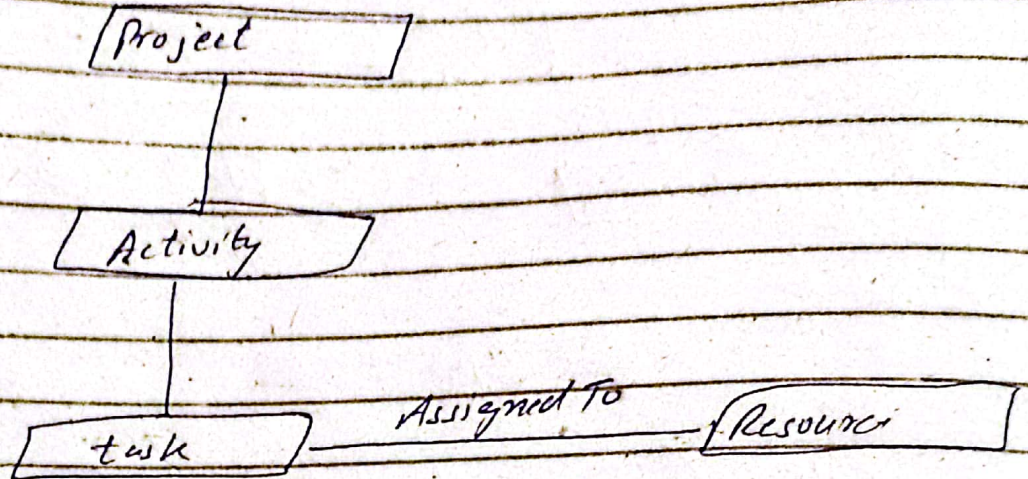
It has members dining table, study table office table.



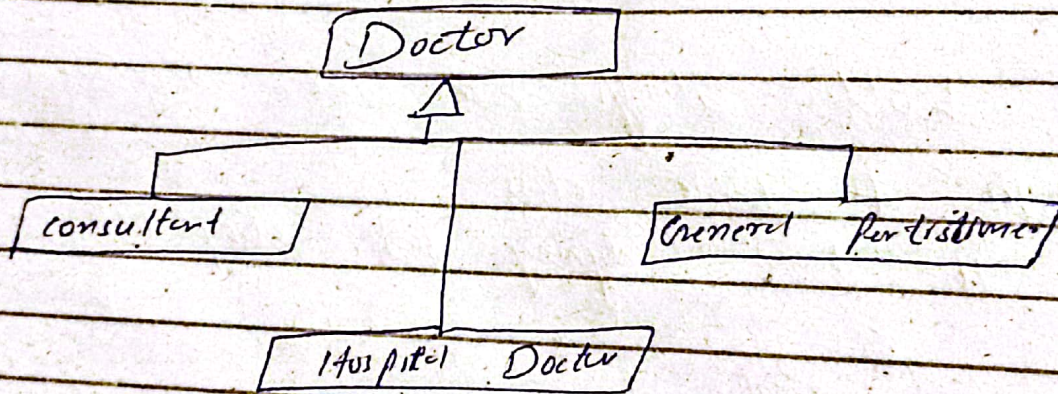
Aggregation/Association



Composition



Generalization



Aggregation

